

tests/dynamic/suites/ concurrency_race_suite.sh — operator guide

Revision: 1 Last modified: 2026-07-01T00:00:00Z Status:
 Authored for P10. Honest-SKIPs (topology_unsupported, §11.4.69)
 until the live dynamic stack lands; parse-clean + authored today.

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview

§11.4.169 **concurrency / race** suite for the external_acl helper. Under heavy concurrent load it asserts: (a) every parallel request gets a **correct, consistent** routing decision (no torn Redis read, no cross-request bleed), (b) **no deadlock/hang** (every request returns within the timeout), (c) **determinism** across the burst (§11.4.50 — identical inputs → identical outcomes). Per-request % {http_code} is captured to its own file (the B3 bluff fix — never a job exit status).

Prerequisites

- Source library tests/dynamic/lib/analyzer_common.sh + committed tests/lib/evidence.sh.
- curl, awk, sort, POSIX sh, date.
- For a real run: HELIX_DYNAMIC_STACK=1 + HELIX_PROXY_URL.
- Optional direct-helper probe: HELIX_ACL_HELPER_CMD — a direct invocation of the Go external_acl binary that reads "<Host>\n" on stdin and prints "OK tag=..." / "ERR".

Usage examples

```
# Today (no live stack): honest SKIP, exit 0:
bash tests/dynamic/suites/concurrency race suite.sh
```

```
# P10 live run with the direct-helper hammer:
HELIX DYNAMIC STACK=1 HELIX PROXY URL=http://127.0.0.1:34128 \
```

```
RACE_CONC=20 HELIX_ACL_HELPER_CMD='./bin/external_acl' \  
bash tests/dynamic/suites/concurrency_race_suite.sh
```

Env: RACE_CONC (default 20), RACE_TARGET (default http://target-a.internal/), RACE_HOST (default target-a.internal).

Edge cases

- **Live stack absent** → topology_unsupported SKIP, exit 0.
- **A hang/deadlock** leaves an empty code file → counted as hung; any hung > 0 fails the GREEN guard.
- **Same Host yields more than one decision** (distinct_tags > 1) under the direct-helper hammer → a race → FAIL.

§11.4.115 RED_MODE polarity

RED_MODE	What it asserts	PASS means
0 (default)	GREEN guard	all 200, 0 hung, ≤1 decision per Host (consistent + live)
1	RED baseline	the pre-fix helper races/hangs/returns inconsistent tags → defect reproduced

A RED_MODE=1 run with no race/hang reproduced is a finding, not a pass.

Internal behaviour

- #!/usr/bin/env bash, POSIX-clean (sh -n + bash -n, §11.4.67).
- Fires RACE_CONC parallel proxied requests → req.N.code; tallies 200s and hung (empty/zero-byte code files).
- When HELIX_ACL_HELPER_CMD is set, also hammers the helper directly in parallel and reduces helper.*.out to sort -u distinct OK tag/ERR decisions; distinct_tags > 1 for one Host = a race.
- Counts + decisions written to concurrency.evidence; evidence dir qa-results/p9-harness/concurrency_race_<run-id>/.

Related

- tests/dynamic/lib/analyzer_common.sh — sourced base.
- tests/dynamic/suites/run_all_suites.sh — orchestrates this suite.

- Constitution §11.4.169 / §11.4.85 / §11.4.50 / §11.4.69 / §11.4.115 / §12.6; design §4/§13.

Last verified

2026-07-01 — run with no live stack: honest SKIP, exit 0; sh -n + bash -n parse-clean. Live concurrent acl-helper drive is exercised in **P10**.